

DATA SCIENCE INTERNSHIP

BEGINNER TASK

Visualization Library Documentation

Matplotlib and Seaborn

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

print("Matplotlib and Seaborn are ready!")
```

Matplotlib and Seaborn are ready!

```
data = {
    "Month": ["Jan", "Feb", "Mar", "Apr", "May"],
    "Sales": [120, 150, 180, 140, 200],
    "Students": [20, 25, 30, 28, 35]
}

df = pd.DataFrame(data)

df
```

	Month	Sales	Students	
0	Jan	120	20	
1	Feb	150	25	
2	Mar	180	30	
3	Apr	140	28	
4	May	200	35	

Next steps: [Generate code with df](#) [New interactive sheet](#)

1. Line Plot

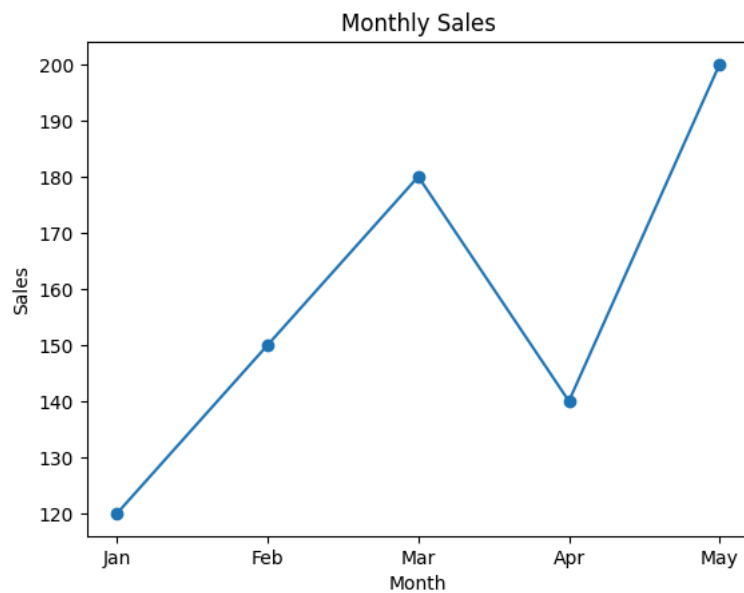
A line plot is used to show changes or trends over time.

Use case: It can be used to show monthly sales trends.

```
plt.plot(df["Month"], df["Sales"], marker="o")

plt.title("Monthly Sales")
plt.xlabel("Month")
plt.ylabel("Sales")

plt.show()
```



2. Bar Chart

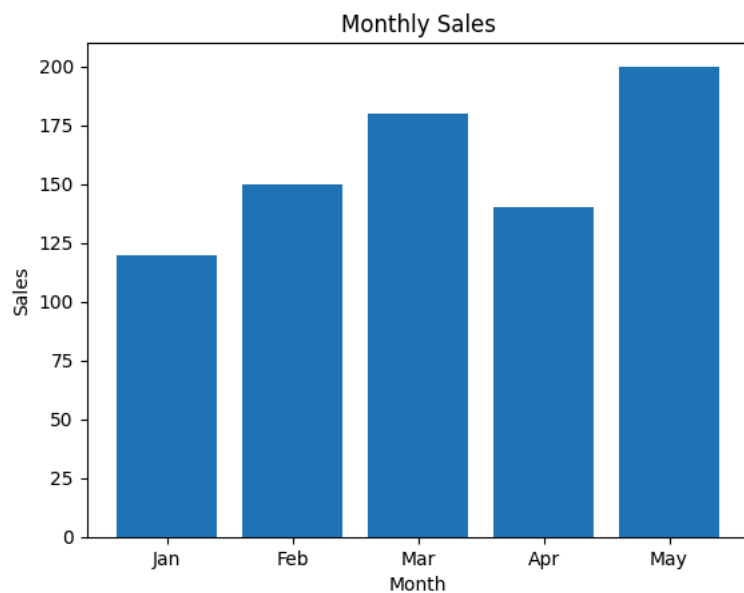
A bar chart is used to compare values between different categories.

Use case: It can be used to compare sales between months.

```
plt.bar(df["Month"], df["Sales"])

plt.title("Monthly Sales")
plt.xlabel("Month")
plt.ylabel("Sales")

plt.show()
```



3. Scatter Plot

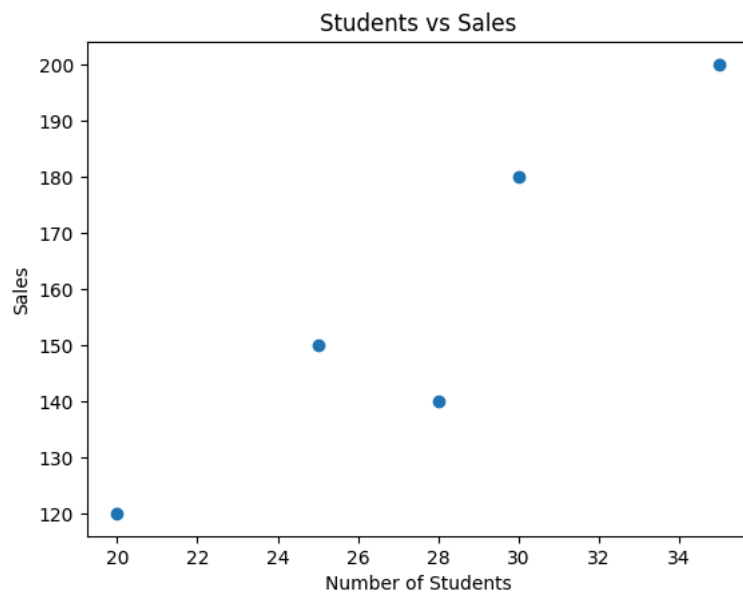
A scatter plot is used to show the relationship between two numerical variables.

Use case: It can be used to study the relationship between students and sales.

```
plt.scatter(df["Students"], df["Sales"])

plt.title("Students vs Sales")
plt.xlabel("Number of Students")
plt.ylabel("Sales")

plt.show()
```



4. Histogram

A histogram is used to show the distribution of numerical data.

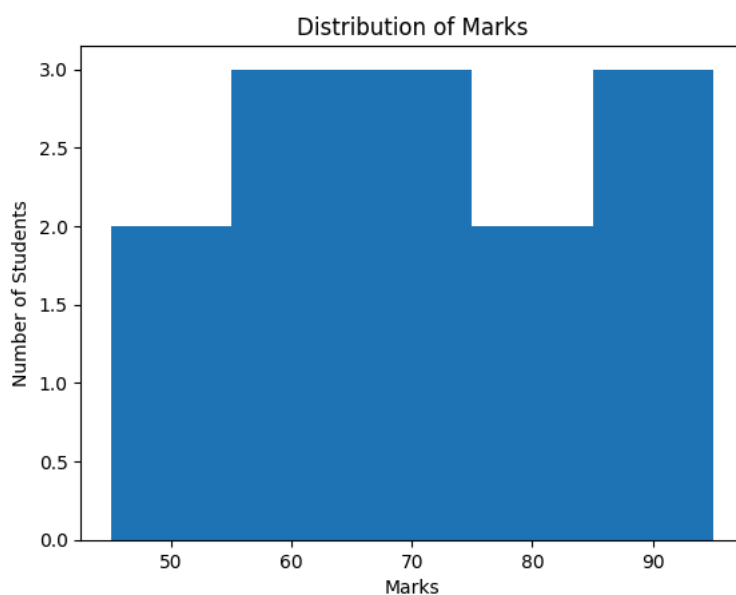
Use case: It can be used to understand how marks are distributed among students.

```
marks = [45, 50, 55, 60, 60, 65, 70, 70, 75, 80, 85, 90, 95]

plt.hist(marks, bins=5)

plt.title("Distribution of Marks")
plt.xlabel("Marks")
plt.ylabel("Number of Students")

plt.show()
```



5. Pie Chart

A pie chart is used to show the proportion or percentage of different categories.

Use case: It can be used to show the distribution of students according to different categories.

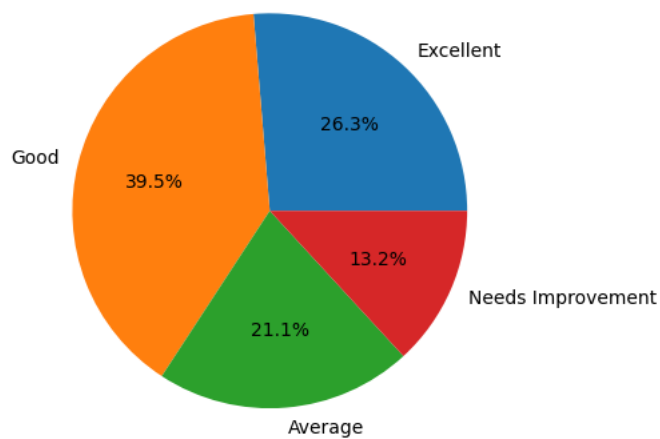
```
categories = ["Excellent", "Good", "Average", "Needs Improvement"]
students = [10, 15, 8, 5]

plt.pie(students, labels=categories, autopct="%1.1f%%")

plt.title("Student Performance Distribution")
```

```
plt.show()
```

Student Performance Distribution



Seaborn

Library Overview

Seaborn is a Python data visualization library built on top of Matplotlib.

It provides simple and attractive statistical graphics and works well with Pandas DataFrames.

Typical Use Cases

Seaborn is commonly used for:

- Statistical data visualization
- Exploring relationships between variables
- Understanding data distributions
- Comparing different categories

1. Line Plot

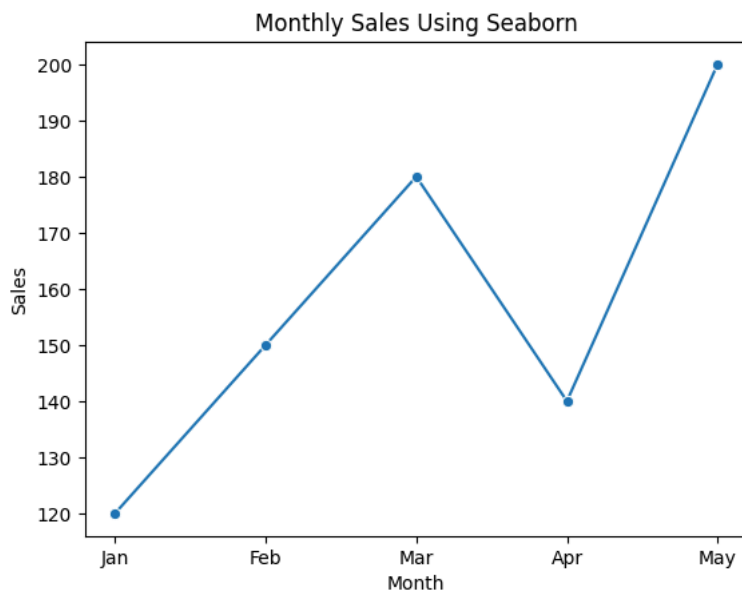
A line plot is used to show trends and changes in data.

Use case: It can be used to show monthly sales trends.

```
sns.lineplot(data=df, x="Month", y="Sales", marker="o")

plt.title("Monthly Sales Using Seaborn")
plt.xlabel("Month")
plt.ylabel("Sales")

plt.show()
```



2. Bar Plot

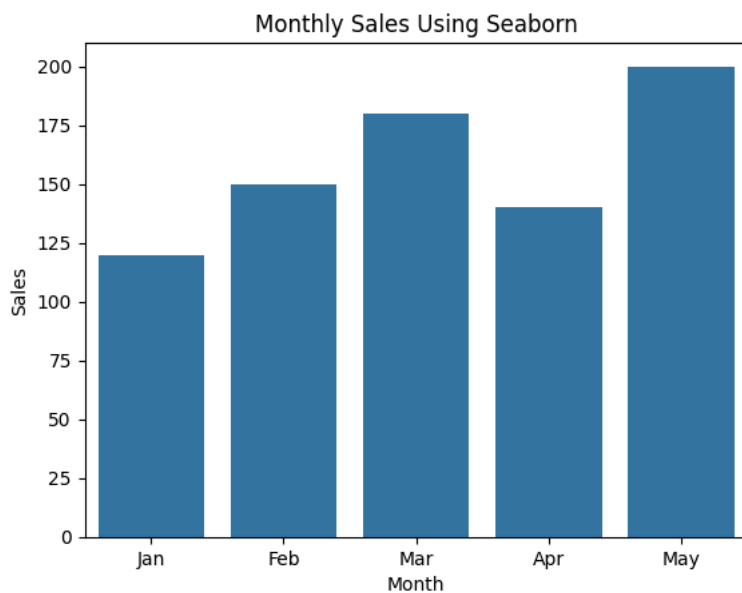
A bar plot is used to compare values between different categories.

Use case: It can be used to compare sales across different months.

```
sns.barplot(data=df, x="Month", y="Sales")

plt.title("Monthly Sales Using Seaborn")
plt.xlabel("Month")
plt.ylabel("Sales")

plt.show()
```



3. Scatter Plot

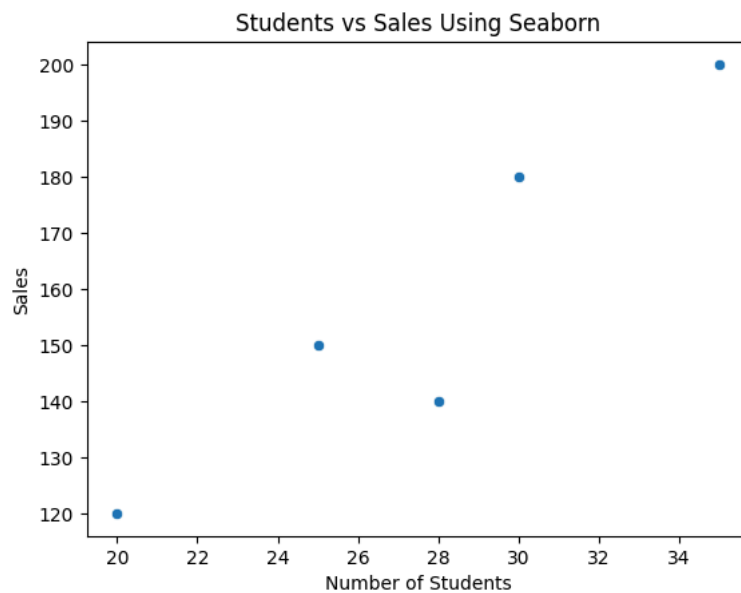
A scatter plot is used to show the relationship between two numerical variables.

Use case: It can be used to study the relationship between the number of students and sales.

```
sns.scatterplot(data=df, x="Students", y="Sales")

plt.title("Students vs Sales Using Seaborn")
plt.xlabel("Number of Students")
plt.ylabel("Sales")

plt.show()
```



Important Parts of the Code

1. `sns.scatterplot()` Creates a scatter plot using Seaborn.
2. `data=df` Uses our Pandas DataFrame.
3. `x="Students"` Uses the Students column for the X-axis.
4. `y="Sales"` Uses the Sales column for the Y-axis.
5. `plt.title()` Adds a title to the graph.
6. `plt.xlabel()` Gives a name to the X-axis.
7. `plt.ylabel()` Gives a name to the Y-axis.
8. `plt.show()` Displays the graph.

4. Histogram

A histogram is used to show the distribution of numerical data.

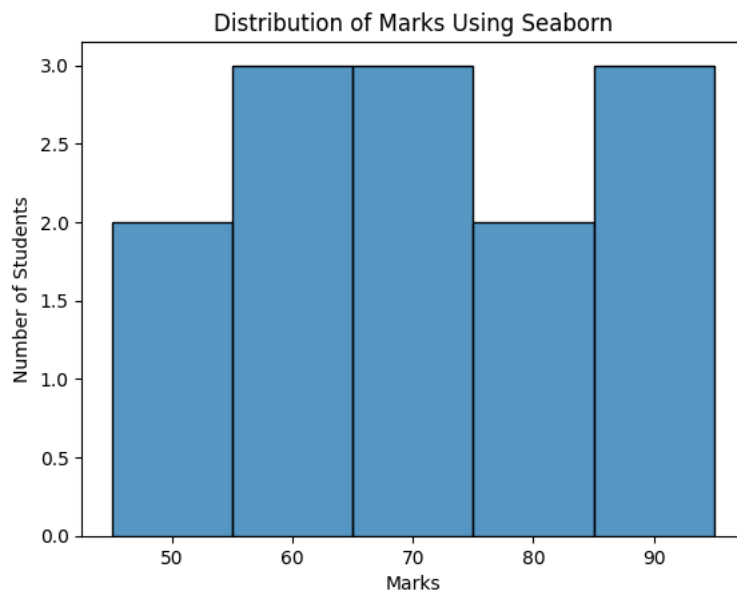
Use case: It can be used to understand how marks are distributed among students.

```
marks = [45, 50, 55, 60, 60, 65, 70, 70, 75, 80, 85, 90, 95]

sns.histplot(marks, bins=5)

plt.title("Distribution of Marks Using Seaborn")
plt.xlabel("Marks")
plt.ylabel("Number of Students")

plt.show()
```



Important Parts of the Code

1. `sns.histplot()` Creates a histogram using Seaborn.
2. `marks` Contains the numerical marks data.
3. `bins=5` Divides the data into 5 groups.
4. `plt.title()` Adds a title to the graph.
5. `plt.xlabel()` Gives a name to the X-axis.
6. `plt.ylabel()` Gives a name to the Y-axis.
7. `plt.show()` Displays the histogram.

5. Box Plot

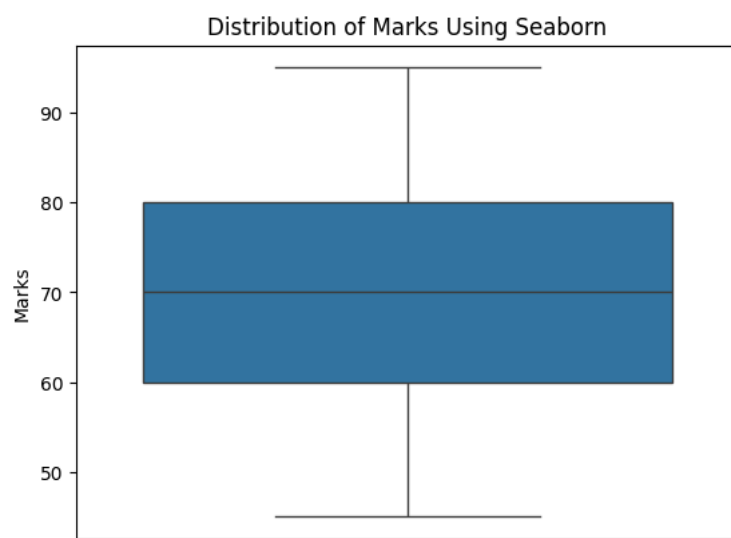
A box plot is used to show the distribution of numerical data and identify possible outliers.

Use case: It can be used to understand the spread of students' marks.

```
sns.boxplot(y=marks)

plt.title("Distribution of Marks Using Seaborn")
plt.ylabel("Marks")

plt.show()
```



Important Parts of the Code

1. `sns.boxplot()` Creates a box plot using Seaborn.
2. `y=marks` Uses the marks data for the box plot.
3. `plt.title()` Adds a title to the graph.
4. `plt.ylabel()` Gives a name to the Y-axis.
5. `plt.show()` Displays the box plot.

Matplotlib vs Seaborn

Comparison

Feature	Matplotlib	Seaborn
Ease of Use	Requires more code and configuration	Easier for statistical plots
Customization	Highly customizable	Good customization
Interactivity	Mainly static plots	Mainly static plots
Statistical Visualization	Basic	Strong support
DataFrame Support	Good	Very good
Large Datasets	Can handle large datasets, but plotting can become slower	Can also handle large datasets, but performance depends on the plot and data size

Matplotlib Strengths

- Highly customizable
- Provides control over almost every part of a graph
- Suitable for creating basic and advanced visualizations

Matplotlib Weaknesses

- Can require more code
- Some graphs need more manual configuration

Seaborn Strengths

- Simple syntax
- Attractive statistical visualizations
- Works well with Pandas DataFrames
- Useful for exploring relationships and distributions

Seaborn Weaknesses

- Provides less low-level control than Matplotlib
- Built mainly for statistical visualization

Conclusion

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